WEB AUTHORING

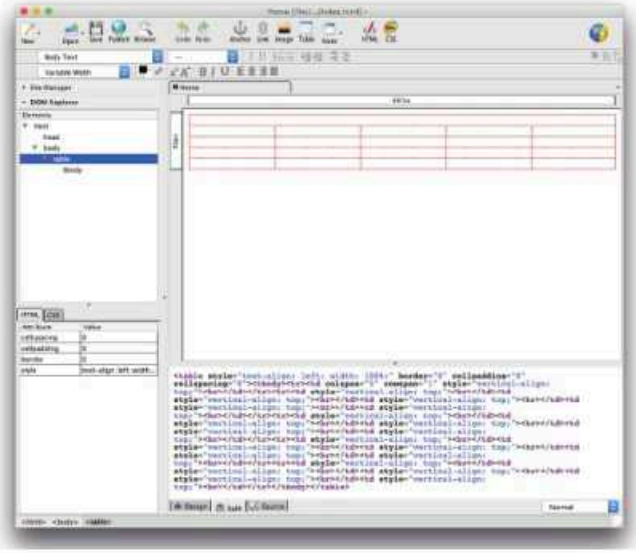
SUBJECT VOCABULARY

Hypertext Markup Language (HTML) a computing language read by web browser software
intranet a computer network used for exchanging or seeing information within a company
web server a computer that stores web pages and sends them to other devices that request them (often using web browser software)

Web authoring software lets users create web pages that include text and images. The pages are output as **Hypertext Markup Language** (HTML). HTML is read by web browser software such as Google® Chrome™[3], Microsoft® Edge or Mozilla® Firefox®[4]. Web browsers translate HTML into pages that people can see and read. Web pages can be linked together to create a website for people to view on an **intranet** or on a **web server** on the internet.

You will learn more about web browsers on page 55 and in *Unit 2 Connectivity* (page 87).

Although some people prefer to create or customise web pages by writing their own HTML, some applications will allow users to produce complex websites with little or no experience of HTML coding. These applications are sometimes called 'What You See Is What You Get' (WYSIWYG) editors, because the published page will look very similar to the page that the user edits in the web authoring application. An example of web authoring software is Adobe® Dreamweaver®[5].



▲ **Figure 2.14** Some web authoring software can display the code and design views one after the other

IMAGE EDITING

SUBJECT VOCABULARY

bitmap a computer image that is stored or printed as an arrangement of bits
vector graphic a graphical image made up of points and lines

Image editing applications let users create and change **bitmap** images, such as digital photographs, or **vector graphics**, such as drawings or logos.

GRAPHICS EDITORS

Graphics editing applications allow users to create or edit vector graphics. An example of graphics editing software is Serif® Affinity Designer[6]. Features of graphics editors include:

- vectorising images (that is, converting or tracing bitmap graphics to vector graphics)
- layers (that is, having the ability to place some graphics on top of others)
- adding text
- adding or drawing shapes and lines
- resizing, aligning or moving shapes and lines
- altering the colour of shapes, lines and fill areas.

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4 MOZILLA FIREFOX IS A TRADEMARK OF THE MOZILLA FOUNDATION

5 ADOBE AND ADOBE INDESIGN ARE EITHER REGISTERED TRADEMARKS OR TRADEMARKS OF ADOBE SYSTEMS INCORPORATED IN THE UNITED STATES AND/OR OTHER COUNTRIES

6 "SERIF" AND "AFFINITY" ARE BOTH REGISTERED TRADEMARKS OF SERIF EUROPE LTD

PHOTO EDITORS

These allow users to edit and enhance digital photographs and images. An example of photo editing software is GNU Image Manipulation Program (GIMP). Features of photo editors include:

- adjusting brightness and contrast
- resizing the image
- altering sharpness and blurring
- applying filters and effects such as distortion
- red-eye removal
- **cropping**.

SUBJECT VOCABULARY

cropping using a photo editor to remove the edges of an image

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
ETHICS
COMMUNICATION
INTERPERSONAL SKILLS

ACTIVITY

▼ EDITING IMAGES

Go to the website of the cosmetics company, Dove, and search for the key terms 'evolution video' to find the video *Evolution*, which is about the way in which the beauty industry uses image editing applications. Watch the video, then discuss the impact that image editing applications can have on society.



▲ **Figure 2.15** There are many photo editing apps available for smartphones; finished images can be stored to local storage or shared online

SOUND EDITING

Sound editing software allows users to edit audio files or to join together different audio files in order to create multitrack music or soundtracks for video. An example of sound editing software is Audacity®. Features of this software include:

SUBJECT VOCABULARY

mute silence an audio track
solo only play one particular audio track
tempo the speed at which a track is played

- cut and join audio clips
- **mute** and **solo** some audio tracks
- alter volume levels for individual tracks
- change **tempo**

GENERAL VOCABULARY

reverberation prolonging a sound

SUBJECT VOCABULARY

normalisation increasing the average volume of a piece of audio to a defined maximum level

- frequency equalisation (changing the levels of high and low pitch frequencies)
- add effects like **reverberation**
- apply audio processing:
  - reverse
  - noise reduction
  - **normalisation**
  - fade the volume in, so that it gets louder, or out, so that it gets quieter.

PRESENTATION SOFTWARE

SUBJECT VOCABULARY

slide a single page of a presentation
pathway the sequence of or route through a series of slides

Presentation software allows users to create engaging multimedia content, including images, text, animation and video. This content can be placed on **slides** or **pathways** that are used to illustrate and support the spoken content of a talk given to an audience. An example of presentation software is Prezi™[7].

Some applications allow users to practise timings and add narration that automatically plays back on the appropriate slides. These slides can be set to advance automatically after a set period of time, or on command from the presenter. Notes can be added to the presentation to remind and prompt the presenter to speak about specific points during the presentation.

CONTROL APPLICATIONS

Control applications are used to make something happen in the physical environment. This type of software is often used to automate the movement of control devices or actuators, such as motors. See page 37 for more information about actuators.

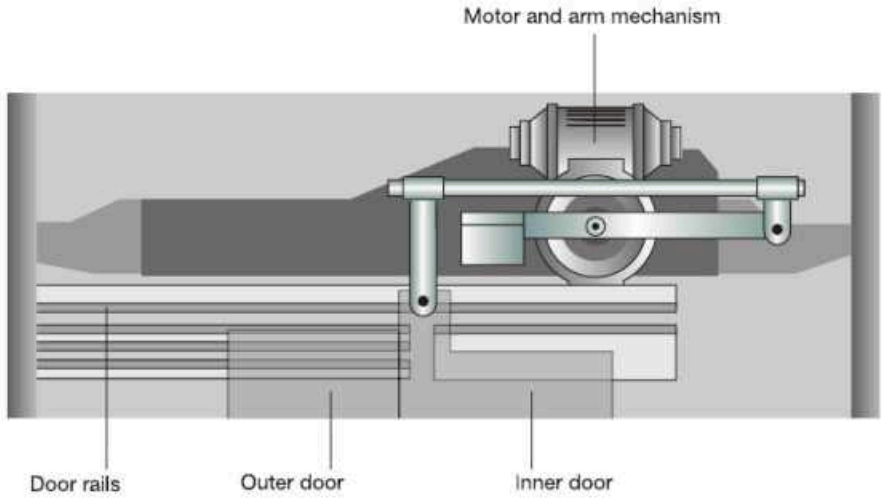




▲ **Figure 2.16** Astronauts on the International Space Station use control software to move a robotic arm into position in order to capture a supply capsule

Control software takes input from one or more sensor(s), makes a decision based on the input value and then outputs something, such as a command, as a result.

7 PREZI IS A TRADEMARK OF PREZI, INC. ALL RIGHTS RESERVED.



▲ **Figure 2.17** Lift doors are opened and closed automatically by control software when a sensor detects that the lift is level with the floor and has stopped

Control software is often used in engineering, vehicles and building control systems. However, a developing group of control applications is in the area of home automation systems, where devices can be monitored and controlled using a smartphone app.



▲ **Figure 2.18** Smartphone apps can be used to control home heating and cooling systems

SKILLS ADAPTIVE LEARNING
INTELLECTUAL INTEREST AND CURIOSITY
COMMUNICATION

ACTIVITY

▼ REMOTE CONTROL

1 Discuss the objects in your home that you think could be connected to the internet in order to be switched on or controlled remotely.

2 Now discuss potential ways in which you could automatically control these connected things in your home based on changes in the environment that are detected by remote sensors or sensors in a smartphone. See Figure 2.19 and the Did you know? feature for some examples.

DID YOU KNOW?

Some smartphone apps can control internet-connected devices in certain situations. For example, if the smartphone calculates that its location is within 10 metres of a garage door, it will send an instruction via the internet to the garage door motors to turn on and open the door.

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
CONTINUOUS LEARNING

ACTIVITY

▼ THE INTERNET OF THINGS

Do some research to find out what is meant by the term, 'the internet of things'.

Control applications are often created by programmers rather than general users and they are not as easy to modify as other application software.

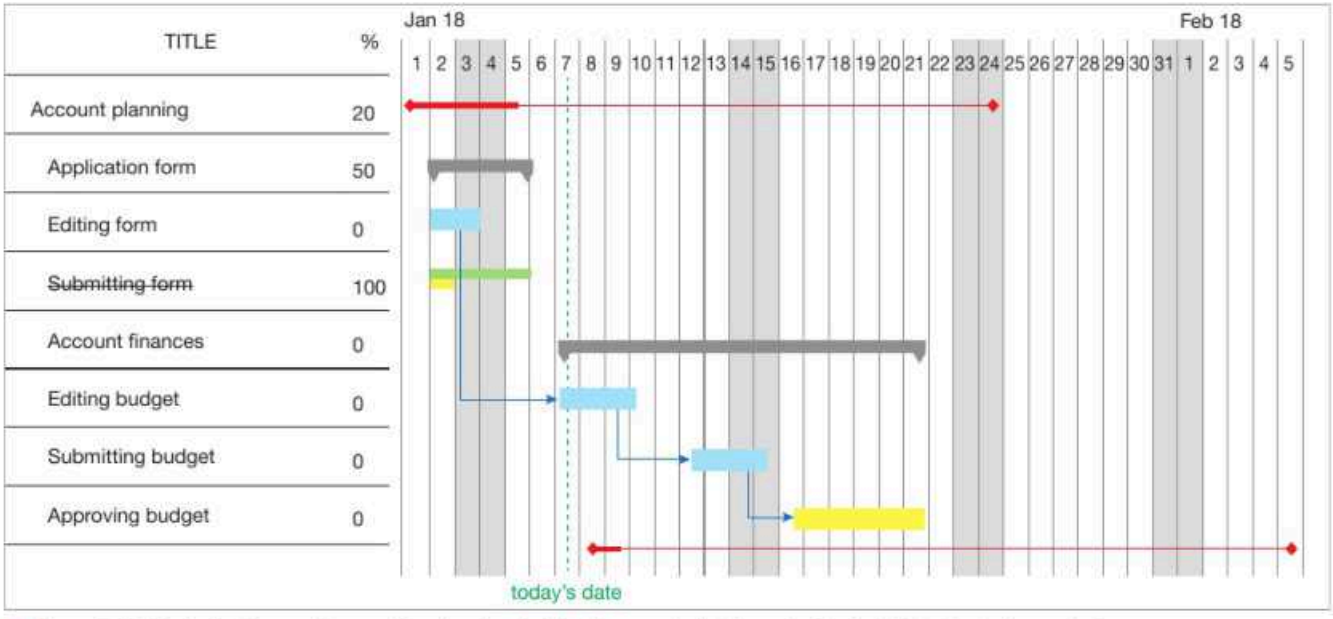
PROJECT MANAGEMENT

GENERAL VOCABULARY

timeline a linear graphical representation of events and the time and order in which they occur
critical path the shortest time a project will take to complete (if no problems or failures are experienced)
milestone a time or date by which a task must be completed

Project management applications are used to help plan and track the individual tasks in a project, so that project managers can make the most efficient use of the available resources. Some tasks cannot be started until a previous task is completed or has been partly completed, so it is important for a project manager to see which tasks are dependent on others. When all of these dependent tasks are combined together into a **timeline**, it is easy for the project manager to see the **critical path**. These applications can also be used to set **milestones**.

Project management applications can be used to allocate tasks to individual people or groups of people. They often provide the option for individuals or groups of people to be allocated to more than one task. Some project management applications also provide tools for tracking the costs and arranging for resources to be delivered on time.



▲ **Figure 2.19** Gantt charts are often used to visualise the timelines and milestones for the individual tasks in a project

COMMUNICATION SOFTWARE

Communication software provides remote access to systems and allows users to contact people using the internet. It can be used to send files and messages as text, images, audio and video.

WEB BROWSERS

Web browsers allow users to view web pages and websites created in web authoring software and hosted on servers that are connected to the internet. These servers may be in another country. Different browsers have different features and compatibility with file types used on the World Wide Web.

Browsers also allow users to access other systems that use the internet but are not on the World Wide Web, such as **file transfer protocol** (FTP) servers.

SUBJECT VOCABULARY

file transfer protocol (FTP) a method of exchanging data and files across a network

EMAIL

Emails can be sent and received by software installed on a computer. This locally installed software acts as a client to an email server (see page 89 for more information about the client-server model). In addition, it often stores or uses a database of contacts, which is sometimes available as another application.

As well as locally installed applications, users can access email through webmail software, which they reach through a web browser. Webmail is an example of a web application that is hosted on a web server and accessed using an internet connection. Some webmail applications allow users to store emails offline for access when they do not have an internet connection.

Email applications often provide features such as labels to help organise emails and **filters**.

SUBJECT VOCABULARY

filter a computer program that only allows certain types of information to pass through it

SOCIAL MEDIA

Social media can be accessed through a web browser or by using locally installed apps.



▲ **Figure 2.20** Social media companies give access to their services through their web pages

Some apps provide access to multiple social media accounts. App developers create apps that people can download and install on their digital devices. TweetDeck is an example of communication software used with social media.



▲ **Figure 2.21** Apps are available for mobile devices as well as PCs

SMS

Short messaging service (SMS) applications are found on mobile phones. They allow users to send up to 160 text characters per message using the mobile phone network. They do not require an internet connection.

MMS

Multimedia messaging service (MMS) applications extend the capabilities of SMS. MMS can deliver more than 160 characters per message and can include video, animations, images and audio. Like SMS, they are sent using the mobile phone network and do not require an internet connection.

INSTANT MESSAGING

Instant messaging applications are very similar to MMS applications, but they require a connection to the internet. They can allow users to see when other users are typing, and users can also prevent others from seeing when they are typing. Messages sent via instant messaging applications can include location data.

SUBJECT VOCABULARY

software licence a legal arrangement that gives a user the right to install and use software

SOFTWARE LICENSING

Sometimes, users require a **software licence** to be able to install and use software on a computer.

There are many types of software licence and the details of software licensing are complicated. To make things easier to understand, you can think about the two types of software that are available:

- free or open-source
- proprietary.

DID YOU KNOW?

Using software without a required licence is called software piracy. Software distributed without the right to do so is called pirate software.

FREE OR OPEN-SOURCE

SUBJECT VOCABULARY

free software software that can be modified or distributed by a user
open-source software software for which the source code is made available to users
source code a collection of instructions that forms a piece of software

Free software licences give users the right to study, modify, copy or distribute a program. The user can decide if, and how much, to charge for a copy of the software or any service provided by the software. This means that free software can be made available for a fee or free of charge. 'Free' refers to the user's freedom to charge whatever they want because there are no restrictions on the use of the software.

Open-source software licences make the **source code** available to users so that they can modify how the software works, or distribute the modified or unmodified software.

HINT

The key difference between the terms 'free' and 'open-source' is that 'free' describes the program and 'open-source' describes the source code.

KEY POINT

In the case of free software, the term 'free' does not refer to whether or not users have to pay for the software. It refers to the fact that the software is free from usage restrictions.

PROPRIETARY

SUBJECT VOCABULARY

proprietary software software that is marketed and distributed by its owner under a brand name
freeware proprietary software provided free of charge to users

Proprietary software is software that is marketed and distributed by its owner under a brand name. The software owner can decide the fee for the software and whether or not the software should be distributed.

Proprietary software can be made available for a fee or free of charge. When no fee is required, the software is called **freeware**. Unlike software distributed under an open-source licence, software with a proprietary licence usually does not make its source code available.

SKILLS ADAPTIVE LEARNING
DECISION MAKING
INTERPRETATION

ACTIVITY

▼ CATEGORISING SOFTWARE

Think about the examples of software types that you have read about in this chapter. Group these examples into the categories 'free or open-source' and 'proprietary'.

SUBJECT VOCABULARY

update (noun) a change or addition to a computer file so that it has the most recent information; (verb) to make a change or addition to a computer file so that it has the most recent information
bugs errors in the program's source code
compatibility ability of one device, system or application to work with other devices, systems and applications
usability the ease with which a device or application can be used

GENERAL VOCABULARY

vulnerability weakness or exposure to harm

SOFTWARE UPDATES

Updates to software are released by software developers for the following reasons:

- fixing security **vulnerabilities** or **bugs**
- increasing **compatibility** with newer operating systems
- improving performance and efficiency
- introducing new features
- improving **usability**.

Software updates are usually made available for download from a server on the internet. Some updates can be scheduled to happen automatically when they are released.

It is important to back up a system and files before updating software in case the update introduces problems, such as removing components that other software needs in order to function.

CHAPTER QUESTIONS

1 State what is meant by the term 'application software'. **(1)**

SKILLS ADAPTIVE LEARNING

2 State **three** types of utility software. **(3)**

SKILLS INTERPRETATION

3 Describe the role of the operating system. **(2)**

SKILLS REASONING

4 Explain why utility software allows the user to schedule backup processes. **(2)**

SKILLS REASONING

5 Explain the benefit of using compression utilities to archive files. **(2)**

SKILLS INTERPRETATION

6 Describe how presentation software can help a presenter to give a talk. **(2)**

SKILLS INTERPRETATION

7 Describe **one** use of control software in a relevant industry. **(2)**

SKILLS CRITICAL THINKING REASONING COMMUNICATION PRODUCTIVITY

8 Discuss why a building company might use project management software. **(8)**

SKILLS PROBLEM SOLVING

9 State **three** types of service provided by communication software. **(3)**

SKILLS REASONING

10 State **two** reasons for updating software. **(2)**